



Week 04: Background Colour with CSS

CSS Colour Options and Backgrounds

Colours in CSS are fundamental in web design, allowing you to enhance the appearance and user experience of your website. CSS supports multiple methods of defining colours, including colour names, RGB values, Hex values, and HSL (Hue, Saturation, Lightness) values.

1. Colour Names

CSS supports a wide range of predefined colour names, such as `red`, `blue`, `green`, `black`, and many more. This is the simplest method to define colours.

```
element {  
  color: blue;  
}
```

This will set the text colour of the element to blue.

2. RGB (Red, Green, Blue)

RGB is another way to define colours by specifying the red, green, and blue components. Each value can range from 0 to 255, where 0 means no intensity and 255 means full intensity.

```
element {  
  color: rgb(255, 0, 0); /* Red */  
}
```

This will set the text colour to red. You can mix different values of red, green, and blue to create a variety of colours.

3. Hexadecimal Colours

Hexadecimal colour codes are another popular method for defining colours. These codes consist of a hash symbol (#) followed by six digits. The first two digits represent the red value, the next two represent green, and the last two represent blue.

```
element {  
  color: #ff0000; /* Red */  
}
```

This will set the text colour to red, similar to the RGB method.

4. HSL (Hue, Saturation, Lightness)

HSL is a more intuitive way of defining colours. It allows you to specify a colour based on its **hue** (the type of colour), **saturation** (the intensity of the colour), and **lightness** (the brightness of the colour).

```
element {  
  color: hsl(0, 100%, 50%); /* Red */  
}
```

This will set the text colour to red, using the HSL notation. The values range from 0 to 360 for hue, and from 0% to 100% for both saturation and lightness.

Applying Background Colours

You can use any of the above colour methods to apply a background colour to an element using the `background-color` property.

```
div {
  background-color: #f0f0f0; /* Light gray background */
  padding: 20px;
}
```

This sets the background colour of the `div` to light gray.

Gradient Colours

Gradients allow you to smoothly transition between two or more colours. There are two types of gradients: linear and radial.

Linear Gradient: A linear gradient transitions between colours in a straight line, either horizontally, vertically, or at an angle.

```
div {
  background: linear-gradient(to right, red, yellow);
  height: 200px;
}
```

This creates a linear gradient that transitions from red on the left to yellow on the right.

Radial Gradient: A radial gradient transitions from a central point outwards in a circular or elliptical shape.

```
div {
  background: radial-gradient(circle, red, yellow);
  height: 200px;
}
```

This creates a radial gradient that transitions from red in the center to yellow at the edges.

Hover Effects with Colours

CSS allows you to change an element's colour when a user hovers over it, making it interactive. You can use the `:hover` pseudo-class to define hover states.

Changing Text Colour on Hover:

```
button {
  background-color: blue;
  color: white;
  padding: 10px;
}

button:hover {
  background-color: green;
  color: black;
}
```

This changes the background colour of a button to green and the text colour to black when the user hovers over it.

Changing Background Colour on Hover:

```
div {
  background-color: lightblue;
  padding: 20px;
}

div:hover {
  background-color: lightcoral;
}
```

This changes the background colour of a `div` to light coral when the user hovers over it.

Opacity and RGBA Colours

If you want to make a colour partially transparent, you can use the `rgba` function. The "a" stands for "alpha," which controls the transparency level. The alpha value ranges from 0 (completely transparent) to 1 (completely opaque).

```
div {
  background-color: rgba(255, 0, 0, 0.5); /* Semi-transparent red */
}
```

```
height: 200px;  
}
```

This applies a semi-transparent red background to the `div` element.

Conclusion

CSS offers a variety of ways to define and use colours. Whether you're using simple colour names, RGB, Hex, or more advanced methods like HSL and gradients, you have plenty of options for enhancing the look and feel of your website. Additionally, CSS provides interactive effects like hover states and opacity changes, which can make your design more engaging for users.



Additional Resources

- [MDN Web Docs - CSS Colours](https://developer.mozilla.org/en-US/docs/Web/CSS/color)  (<https://developer.mozilla.org/en-US/docs/Web/CSS/color>)
- [W3Schools - CSS Colours](https://www.w3schools.com/css/css_colors.asp)  (https://www.w3schools.com/css/css_colors.asp)

